

Electricity Bill Generator

1. Problem Statement

Write a **C program** to generate an electricity bill based on the **number of units consumed** by a customer. The program should accept the number of units consumed as input and calculate the total electricity bill using the given slab-wise tariff. The electricity charge should be calculated separately for each applicable slab.

2. Electricity Tariff

Units Consumed	Rate per Unit
0 – 100	■1.50
101 – 200	■2.50
201 – 500	■4.00
Above 500	■6.00

3. Input

Number of units consumed

4. Output

Total Electricity Bill

5. Calculation Rules

Case 1: Units ≤ 100

$$\text{Bill} = \text{Units} \times 1.50$$

Case 2: Units between 101 and 200

$$\text{Bill} = (100 \times 1.50) + ((\text{Units} - 100) \times 2.50)$$

Case 3: Units between 201 and 500

$$\begin{aligned}\text{Bill} &= (100 \times 1.50) \\ &+ (100 \times 2.50) \\ &+ ((\text{Units} - 200) \times 4.00)\end{aligned}$$

Case 4: Units > 500

$$\begin{aligned}\text{Bill} &= (100 \times 1.50) \\ &+ (100 \times 2.50) \\ &+ (300 \times 4.00) \\ &+ ((\text{Units} - 500) \times 6.00)\end{aligned}$$

6. Algorithm

Step 1 — Start the program.

Step 2 — Declare the variables units and bill.

Step 3 — Read the number of units consumed from the user.

Step 4 — Check whether units are less than or equal to 100. If yes, calculate $\text{bill} = \text{units} \times 1.50$.

Step 5 — Otherwise, check whether units are less than or equal to 200. If yes, calculate $\text{bill} = (100 \times 1.50) + ((\text{units} - 100) \times 2.50)$.

Step 6 — Otherwise, check whether units are less than or equal to 500. If yes, calculate $\text{bill} = (100 \times 1.50) + (100 \times 2.50) + ((\text{units} - 200) \times 4.00)$.

Step 7 — If none of the above conditions are satisfied, calculate $\text{bill} = (100 \times 1.50) + (100 \times 2.50) + (300 \times 4.00) + ((\text{units} - 500) \times 6.00)$.

Step 8 — Display the calculated electricity bill.

Step 9 — Stop the program.

7. Pseudocode

BEGIN

 DECLARE units AS INTEGER

 DECLARE bill AS FLOAT

 SET bill = 0

 INPUT units

 IF units <= 100 THEN

 bill = units * 1.50

 ELSE IF units <= 200 THEN

 bill = (100 * 1.50) +
 ((units - 100) * 2.50)

 ELSE IF units <= 500 THEN

 bill = (100 * 1.50) +
 (100 * 2.50) +
 ((units - 200) * 4.00)

 ELSE

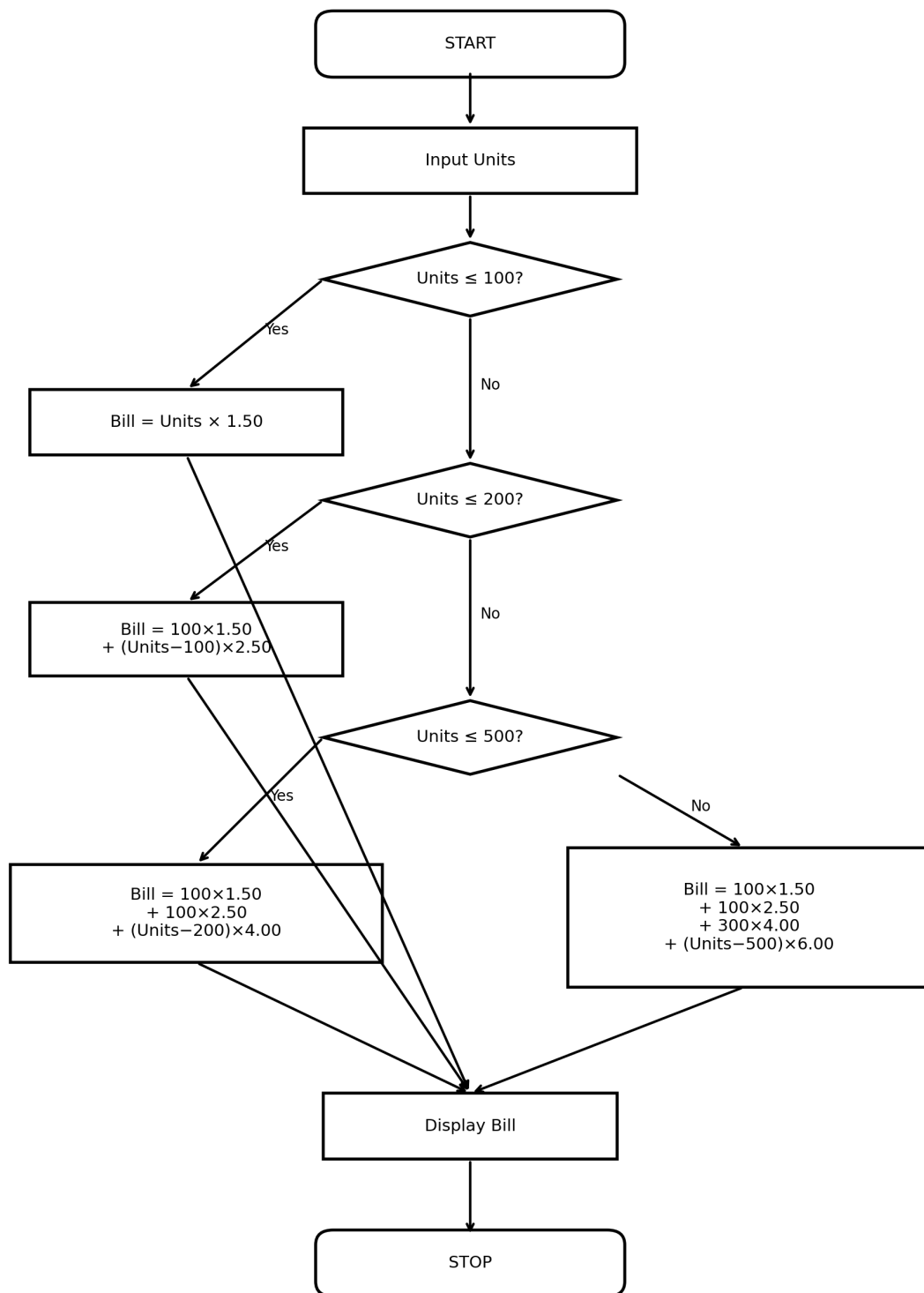
 bill = (100 * 1.50) +
 (100 * 2.50) +
 (300 * 4.00) +
 ((units - 500) * 6.00)

 END IF

 DISPLAY bill

END

8. Block Diagram / Flowchart



9. Sample Test Cases

Test Case	Units	Expected Bill
1	0	■0.00
2	50	■75.00
3	100	■150.00

Test Case	Units	Expected Bill
4	150	■275.00
5	200	■400.00
6	250	■600.00
7	350	■1,000.00
8	500	■1,600.00
9	600	■2,200.00
10	750	■3,100.00

10. Detailed Example

Input:

Enter units consumed: 350

First 100 units

= $100 \times \text{■}1.50 = \text{■}150$

Next 100 units

= $100 \times \text{■}2.50 = \text{■}250$

Remaining 150 units

= $150 \times \text{■}4.00 = \text{■}600$

Total Bill

= $\text{■}150 + \text{■}250 + \text{■}600$

= $\text{■}1,000$

Output:

Total Electricity Bill = $\text{■}1000.00$

11. C Program

```
#include <stdio.h>
```

```
int main() {
    int units;
    float bill = 0;

    printf("Enter units consumed: ");
    scanf("%d", &units);

    if (units <= 100) {
        bill = units * 1.50;
    }
    else if (units <= 200) {
        bill = (100 * 1.50) +
            ((units - 100) * 2.50);
    }
    else if (units <= 500) {
```

```
        bill = (100 * 1.50) +  
                (100 * 2.50) +  
                ((units - 200) * 4.00);  
    }  
    else {  
        bill = (100 * 1.50) +  
                (100 * 2.50) +  
                (300 * 4.00) +  
                ((units - 500) * 6.00);  
    }  
  
    printf("Total Electricity Bill = Rs. %.2f\n", bill);  
  
    return 0;  
}
```

12. Sample Execution

```
Enter units consumed: 350  
Total Electricity Bill = Rs. 1000.00
```

Key Concept: This problem demonstrates input/output, variables and data types, arithmetic operators, if-else-if conditions, slab-wise calculation, and formatted output.